

TK-Shield — Full Project Report

Defensive bio-piracy monitoring & novelty assessment for Traditional Knowledge

Author: Ansul Suryawanshi - Status: Working, verified end-to-end - Licence: MIT

A companion to the concise [project brief PDF](TK-Shield-Whitepaper.pdf). This report tells the complete story: what TK-Shield is, why it matters, how it is built, the decisions behind it, how it was evaluated and hardened, and where it can go next. It is written to be read by both policy reviewers (WIPO IGC, IP offices) and technical reviewers (engineers, interviewers).

Executive summary

Traditional Knowledge (TK) — documented practices such as *turmeric for wound healing* or *neem as a crop antifungal* — is routinely misappropriated by patents that claim it as novel. Correcting such a patent after grant is slow and expensive, and the communities who hold the knowledge rarely have the tooling to act.

TK-Shield is a defensive platform that answers one question for those communities, the NGOs that support them, and the patent offices that examine applications:

"Has someone patented our traditional knowledge — and can we prove prior art to stop them?"

Given a documented practice, TK-Shield finds the patents that may claim it, scores bio-piracy risk on an interpretable 0–100 scale, gathers citable prior-art evidence from free public databases, and uses a local language model to draft a citation-backed assessment plus a patent-opposition document. An examiner-facing mode runs the inverse: paste a patent and get a **claim-by-claim** novelty verdict against documented TK.

It runs **keyless and offline-first** — the entire pipeline works with zero API keys, on public-domain data and a local model, so it can run on a community laptop with no recurring cost or vendor lock-in.

Headline facts

Patent corpus	16,292 real US patents (PatentsView bulk — real titles, assignees, dates)
TK registry	2,030 documented practices (Dr. Duke CC0 + curated multilingual Wikidata)
Landmark evaluation	Precision@1 = 100%, Precision@5 = 100%, MRR = 1.000, 0% control false positives
Tests	121 backend (network-free) + 22 frontend, green in CI
Cost to run	Zero — no API keys, no paid services, local LLM optional

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1. The problem — bio-piracy and traditional knowledge

Bio-piracy is the appropriation of traditional or indigenous knowledge of nature — typically medicinal or agricultural uses of plants — by an entity that patents it without recognition, consent, or benefit-sharing. The knowledge is often centuries old and orally transmitted, so it rarely appears in the patent literature an examiner searches. The result is a patent granted over something that was never novel, but whose prior art is invisible to the system.

The field is defined by three landmark disputes, each eventually overturned using documented prior art — and each seeded into TK-Shield so the tool always demonstrates them:

Case	Patent	Claimant	Outcome
Turmeric for wound healing	US5401504A	University of Mississippi Medical Center	Revoked by the USPTO in 1997 on documented Indian prior art
Neem as a crop antifungal	EP0436257B1	W.R. Grace & Co. / USDA	Revoked by the EPO in 2000/2005 on documented Indian prior art
Basmati aromatic rice	US5663484A	RiceTec Inc.	Most claims withdrawn/struck at the USPTO in 2001–2002

These cases took years and significant legal resource to reverse. The premise of TK-Shield is that **the prior art existed and was findable** — the gap was tooling that could connect a documented practice to the patents claiming it and assemble the evidence quickly. India's **Traditional Knowledge Digital Library (TKDL)** proved the defensive-documentation model works at the state level; TK-Shield

mirrors that purpose using open data and open models so it is reproducible by anyone.

2. Policy context — alignment with WIPO's mandate

TK-Shield operationalizes the **defensive protection** of traditional knowledge that international IP policy is converging on:

- **WIPO IGC** — the Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge and Folklore. Its core agenda is preventing the *erroneous granting* of patents over TK. TK-Shield's examiner mode is a direct instrument of that goal.
 - ****WIPO Treaty on IP, Genetic Resources and Associated Traditional Knowledge (2024)**** — introduces a *disclosure-of-origin* requirement for patents based on genetic resources and associated TK. TK-Shield supports compliance: it links a practice to the patents claiming it and assembles the origin evidence.
 - **Nagoya Protocol / Convention on Biological Diversity** — access and benefit-sharing depends on knowing *which community* holds a practice. TK-Shield surfaces documented **communities and peoples** (e.g. Santal, Seri, Kwakiutl) as a first-class attribution dimension, not an afterthought.
 - **TKDL** — the proven national model TK-Shield generalizes with open data. The design rule that ties this together: ****every risk claim ties to a stable, verifiable reference**** — a PubMed PMID, a Wikidata QID, a GBIF key, or a patent number. The output is evidence, not assertion.
-

3. What TK-Shield does — three personas, one platform

TK-Shield serves three real audiences through one platform (no auth/workflow layer — a deliberate scope choice for a demonstration tool):

Defender (communities / NGOs)

Register a documented practice -> run a fast risk check -> generate a full RAG report (assessment + opposition draft) -> monitor newly-filed patents -> export. The risk recommendations point to *real channels*: WIPO PATENTSCOPE, the TKDL, and the WIPO IGC.

Examiner (patent offices)

Paste a patent's text and get a novelty verdict — ****LIKELY NOT NOVEL / POSSIBLE PRIOR ART / LIKELY NOVEL**** — against the documented TK registry, with the matching prior art. When claims are present, TK-Shield assesses novelty ****claim by claim****: it splits the patent into its individual independent/dependent claims and verdicts each against documented TK, the way an examiner checks anticipation. The

verdict is computed from cosine similarity, **not** the LLM, so it is immune to prompt injection in the pasted text.

Researcher

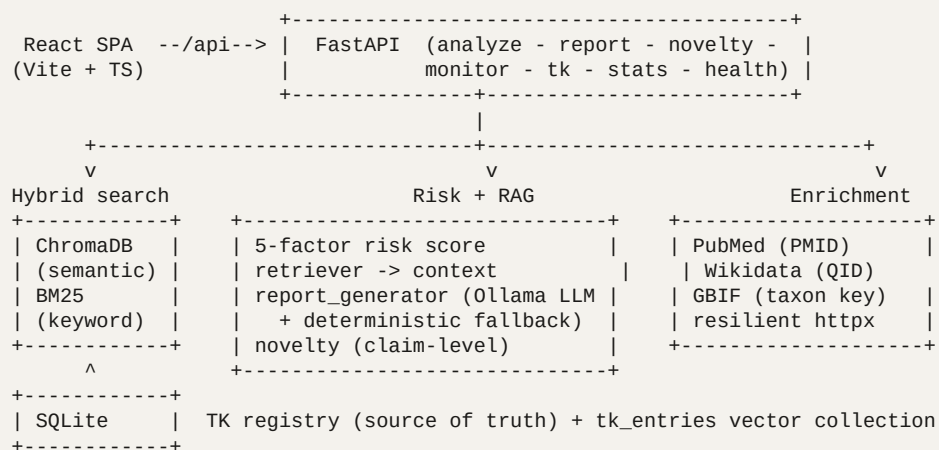
Aggregate analytics across the registry and the corpus: distribution by domain, geography, documented **communities and peoples**, and top assignees.

The end-to-end flow:

```
TK entry (registry) -> hybrid search over patents (RRF: semantic 0.7 + BM25 0.3)
-> 5-factor risk score -> prior-art enrichment (PubMed / Wikidata / GBIF)
-> RAG report via local LLM (assessment + opposition draft)
-> FastAPI endpoints -> React dashboard
    (+ optional live PatentsView monitoring of newly-filed patents)
```

4. System architecture

TK-Shield is a Python backend (FastAPI) over two vector collections and a SQLite registry, with a React single-page frontend served from the same origin.



Module layout (absolute `src.* / api.* imports`, run as `python -m ...`):

- `src/search/` — `vector_store` (ChromaDB), `keyword_search` (BM25), `hybrid_ranker` (RRF fusion + kind-code dedup). The main search path.
- `src/classifier/` — `ip_risk_scorer` (5-factor model), `domain` (medicinal / agricultural / food / cosmetic tagging).
- `src/nlp/` — `preprocessor` (clean/tokenize/lemmatize), `ner_extractor` (dictionary NER for plants/uses/practices + transliterations, spaCy for places).
- `src/clients/` — resilient keyless API wrappers (PubMed, Wikidata, GBIF, PatentsView). Shared `_http` retries, times out, and `**returns None on failure` — never raises**.
- `src/enrichment/prior_art` — concurrent fan-out to the clients -> one deduped, citation-tagged evidence bundle with `sources_used / sources_skipped`.
- `src/rag/` — `retriever` (assembles context), `report_generator` (assessment + opposition; LLM with deterministic fallback), `novelty`

(examiner reverse lookup, claim-level), `claim_parser`, `llm_client` (Ollama).

- `src/registry/tk_store` — CRUD over SQLite + the `tk_entries` vector collection.
- `src/ingestion/` — pluggable patent sources and TK sources, `corpus/registry` builders, ChromaDB indexer, landmark seeder, demo seeder.
- `api/` — app, cached singletons (deps), request schemas, and routers.

A core convention keeps the data layer consistent: a **CSV is the source of truth** for patents; the ChromaDB patents collection and the in-memory BM25 index are both built from that same CSV via one loader, so the three stay **1:1** (verified 16,292 each). Storage access is isolated behind `vector_store` / `tk_store` so a future database swap stays localized.

5. How it works — the technical core

5.1 Hybrid retrieval (semantic + keyword, fused by RRF)

A single retrieval method has blind spots: pure keyword search (BM25) misses folk and multilingual synonyms ("haldi" for turmeric, "curcumin" for the active compound); pure semantic search can drift on exact technical terms. TK-Shield runs **both** and fuses them with **Reciprocal Rank Fusion (RRF)**:

$$\text{score}(\text{doc}) = \frac{\text{weight}}{(k + \text{rank})} \quad k = 60$$

- Semantic side: all-MiniLM-L6-v2 sentence embeddings (384-dim) in ChromaDB, cosine distance, $\text{similarity} = 1 - \text{distance}$.
- Keyword side: in-memory BM25 (`rank-bm25`).
- Weights: **semantic 0.7, keyword 0.3** (config-tunable).

RRF is rank-based, not score-based, so it combines the two systems without having to calibrate their incomparable score scales. The ranker also **de-duplicates** kind-code variants** (US5401504 vs US5401504A) by normalizing the patent id, merging metadata so a richer record backfills empty fields on its twin — which matters because a seeded landmark row may carry an IPC code the bulk row lacks.

5.2 The 5-factor risk model

Risk is an **interpretable, hand-weighted** 0–100 score — deliberately not a black-box classifier, because the output must be explainable to a community or a patent office. Five factors, each capped:

Factor	Max	What it measures
Similarity	40	How close is the nearest patent? (continuous, monotonic curve)
Temporal	20	Was a candidate patent filed after the TK was documented?
Geographic	15	Are candidate patents filed in a country other than the TK origin?

Factor	Max	What it measures
Assignee	15	Corporate / known-actor assignee vs. academic vs. individual
IPC class	10	Is a candidate in a historically bio-piracy-prone patent class?

Bands: **CRITICAL** ≥ 80 , **HIGH** ≥ 60 , **MEDIUM** ≥ 40 , **LOW** ≥ 20 , else **MINIMAL**.

Two specificity rules prevent false positives — the failure mode that would destroy trust in a defensive tool:

- **Relevance gate.** The four "aggravating" factors (temporal, geographic, assignee, IPC) only apply when there is a **credible** candidate — top similarity ≥ 0.50 . Below the gate, risk reflects similarity alone. This stops a benign practice that only weakly matches the corpus from being inflated by structural traits (foreign filing, corporate assignee) that are meaningless without a real match.
- **Missing data is not risk.** An unknown date / country / assignee / IPC contributes **0**, never an "assume risk" default. The model scores evidence, not gaps.

The response surfaces `relevance_gated` for transparency, plus the full factor breakdown and a fixed list of channel-specific recommendations per band.

5.3 Prior-art enrichment

For the plants and uses in a TK entry, TK-Shield fans out **concurrently** to three keyless databases and returns one normalized bundle:

- **PubMed** (NCBI E-utilities) -> peer-reviewed literature, cited by **PMID**.
- **Wikidata** -> taxonomy + multilingual aliases, cited by **QID**.
- **GBIF** -> species occurrence / native range, cited by **taxon key**.

Each item carries its source and a stable id, deduped across common/scientific names. Every source degrades independently: if one is disabled, empty, or down it is recorded in `sources_skipped` and the rest proceed. Nothing here can crash the pipeline.

5.4 RAG report generation

The retriever assembles candidate patents + risk + evidence into a context, and `report_generator` prompts a **local LLM (Ollama, llama3.2)** for two fields: a markdown **assessment** and a formal **opposition draft**. Three guarantees make the output trustworthy:

- **Deterministic fallback.** With no LLM running, a templated report is produced instead — figures and citations stay exact; only the prose is templated. The tool is fully usable offline.
- **No hallucinated citations.** Any reference id the LLM narrative mentions that

is **not** in the verified citation/patent set is surfaced as

unverified_citation_refs — a trust guard against fabricated evidence.

- **Exact numbers.** The risk score, factor breakdown, and citation list are always computed deterministically; the LLM only writes narrative around them.

5.5 Claim-level novelty (the examiner core)

The examiner flow accepts up to 50,000 characters of pasted patent text. When that text contains claim structure, `claim_parser` splits it into individual claims —

detecting numbering and independent vs. dependent relationships ("The method of claim 1, wherein..."). Each claim is searched independently against the TK registry, producing a **per-claim verdict** from cosine similarity (thresholds:

NOT-NOVEL ≥ 0.60 , POSSIBLE ≥ 0.45). The overall verdict aggregates like an examiner does: a patent fails novelty if **any independent claim** is anticipated.

If no claim structure is present (e.g. only an abstract was pasted), it falls back to whole-text assessment — the contract is uniform.

Critically, **the verdict is computed from similarity, not the LLM** — so a patent applicant cannot manipulate the verdict by embedding instructions in the patent text. The LLM only narrates the already-decided verdicts.

6. Key design decisions and trade-offs

Decision	Rationale	Trade-off accepted
Keyless & offline-first	Any community can run it; no cost, no vendor lock-in, no deprecation risk	Smaller local model; one-time bulk data download
Hybrid retrieval over pure embeddings	Recovers folk/multilingual/scientific synonyms a single method misses	More moving parts than a single index
Interpretable risk model (not learned)	Output must be explainable to communities and offices; no labelled training set exists	Hand-weighted, not data-optimized
Deterministic fallbacks everywhere	Graceful degradation is a hard requirement for a tool used in low-resource settings	More code paths to maintain and test
Similarity-computed verdicts	Immune to prompt injection in untrusted patent text	LLM cannot override an obviously-wrong threshold call
Real-metadata corpus only	Credibility — synthetic rows are excluded from the index	Bounded corpus size vs. an inflated synthetic one
Three personas, no auth	Demonstrates the full value to all stakeholders without a workflow layer	Not multi-tenant; single-user/local

The throughline: ****every choice favours trustworthiness and reproducibility over raw capability****, because a defensive IP tool that is occasionally wrong, opaque, or unaffordable is worse than no tool.

7. Data sources and corpus integrity

All sources are keyless and verified live:

Source	Role	Licence
PatentsView bulk (USPTO)	Patent corpus — real titles, assignees, grant dates	Public domain
Dr. Duke's Phytochemical & Ethnobotanical DB (USDA)	TK registry ethnobotany	CC0
Wikidata	Curated multilingual TK seed + taxonomy aliases	CC0
PubMed (NCBI E-utilities)	Literature prior art	Public
GBIF	Species / native-range evidence	Open
Ollama (llama3.2)	Local report narrative	Local, optional

Corpus integrity is enforced, not assumed:

- A strict TK-relevance keyword filter and an **exclusion of synthetic sources**

(a low-fidelity HuggingFace fallback used only for offline bootstrap) mean the searchable corpus is **100% real metadata**.

- The same loader feeds the ChromaDB rebuild and the API's BM25, so

CSV ? ChromaDB ? BM25 stay 1:1 (verified 16,292 each).

- **NaN hygiene:** empty CSV/TSV cells (read by pandas as NaN) are cleaned to empty strings throughout ingest, so the literal string "nan" is never stored or displayed; the frontend shows "—" for empty values.

- A migration splits Duke's COUNTRY(PEOPLE) labels into a clean country + a community field for Nagoya/WIPO-IGC attribution, with a country/region stoplist so a country is never mis-stored as a holder people.

8. Evaluation and results

The evaluation harness (src/evaluation/landmark_eval.py) submits

independently-authored, folk-worded TK descriptions — different sentences from the patents, though sharing the plant/use terms a real registrant would use — for the three landmark cases through the *full* pipeline, and measures retrieval and risk.

Retrieval and risk (corpus of 16,292 real US patents)

Case	Expected patent	Rank	Similarity	Risk
Turmeric for wound healing	US5401504A	1	0.823	CRITICAL (86)
Neem as crop antifungal	EP0436257B1	1	0.787	CRITICAL (94)
Basmati aromatic rice	US5663484A	1	0.662	CRITICAL (83)

Metric	Result
Precision@1	100%
Precision@5	100%
Mean Reciprocal Rank	1.000
Cases flagged HIGH/CRITICAL	100%

Metric	Result
Control false-positive rate	0%

Three benign control practices — "drinking warm water in the morning", "afternoon walk for relaxation", and "distributed cloud job scheduler" — were all correctly left at **MINIMAL** (scores 12, 9, 4), relevance-gated, matching no landmark. This is the false-positive guard working.

Honesty about the ablation

A retrieval ablation (BM25-only vs. semantic-only vs. hybrid) shows all three methods rank the target at #1 on these canonical cases — because the shared plant terms let keyword search alone already find them. **The report says so plainly.**

Hybrid's added value is for the synonym/multilingual queries these specific descriptions do not stress. This is a ****small demonstration on canonical cases, not a population-scale benchmark**** — and stating that clearly is itself a credibility feature. The honest framing is repeated in the README and the brief.

Reproduce: `PYTHONPATH=. python -m src.evaluation.landmark_eval.`

9. Security and hardening

TK-Shield underwent a structured engineering-board audit; the findings were fixed in a dedicated hardening pass (merged as PR #9). Security is treated as a property of construction, not a later add-on.

Confirmed-and-fixed vulnerability — path traversal (the headline finding).

The SPA catch-all route served any file reachable from `dist / <path>`. Starlette decodes `%2e%2e -> ..` and does not normalize the path parameter, so a request for `/%2e%2e/%2e%2e/api/main.py` ****leaked source code, .env, and the SQLite registry**** — verified through the ASGI stack. The fix requires the resolved path to stay under the build directory (`is_relative_to`), with a regression test. Live re-verification: the same request now returns the SPA shell; source is not served.

Other hardening:

- **Per-IP rate limiting** (slowapi, in-memory, keyless) via per-route decorators — never throttling static assets, the SPA, or `/api/health` — with a tighter cap on the expensive LLM routes (`/report`, `/novelty`), which are ***also*** bounded by a concurrency semaphore (excess returns a clean 503).
- **Configurable CORS** (default `*` for local; tighten via env for deployment).
- **Bounded inputs** — every request field is length-capped so a multi-megabyte paste or a huge `n_results` never reaches embeddings, the LLM, or SQLite.
- **XSS-safe by construction** — server/LLM/user text is rendered through `react-markdown` with **no raw-HTML pass-through**; citation hrefs are

scheme-validated (http/https only), blocking javascript:/data: URIs. There are dedicated frontend XSS-safety regression tests.

- **Prompt-injection-immune verdicts** — novelty verdicts are similarity-computed, so untrusted patent text cannot steer them.
 - **SQLite robustness** — WAL mode + busy-timeout so concurrent threadpool handlers and bulk writes don't hit "database is locked"; LIKE metacharacters in search are escaped (still bound parameters — no injection).
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10. Quality engineering

- **Tests** — **121 backend** tests (fully network-free; external APIs and the LLM are mocked/fixed) and **22 frontend** tests (Vitest, including the XSS-safety regressions). A `pyproject.toml` puts the repo root on the path so the documented `pytest tests/` works directly.
 - **Continuous integration** — GitHub Actions runs the backend suite and the frontend typecheck + tests on every push and PR; both PRs in the hardening and feature work merged green.
 - **One-command demo** — a multi-stage Dockerfile + compose file boot an instant landmark-case demo (`docker compose up --build -> http://localhost:8000`) with the three cases seeded and **no 219 MB corpus download**.
 - **Reproducibility** — fully pinned `requirements.txt` and a `package-lock.json`; no runtime CDN; every `src/` module keeps a `__main__` smoke block.
 - **Evaluation as code** — the brief PDF is rendered from the eval JSON, so the published numbers can never drift from the measured ones.
-

11. Technology stack

Backend — Python 3.11 - FastAPI - ChromaDB (vector search) - rank-bm25 (keyword) - SQLite (registry) - sentence-transformers (all-MiniLM-L6-v2) - spaCy NER - resilient httpx clients - slowapi (rate limiting) - Ollama (llama3.2, local LLM) - reportlab (PDF) - loguru.

Frontend — Vite - React 18 - TypeScript - React Router - TanStack Query - Tailwind v4 - Radix primitives - Lucide icons - self-hosted Inter font - Vitest.

Ops — Docker + docker-compose - GitHub Actions CI - .env-driven config (a single config singleton; every value has a sensible default).

12. Limitations

Stated honestly — TK-Shield is ****decision-support that flags candidates for expert human review; it does not establish or prove misappropriation.****

- **Scope** — US patent *metadata* (titles, abstracts, assignees, dates) for the corpus; non-US full text is not yet covered. (The examiner mode does parse claims from pasted text.)
 - **Language** — English-centric; multilingual TK names are matched but the pipeline is English-first.
 - **Model** — the local LLM is small (llama3.2); narrative quality is below frontier API models — a deliberate keyless/offline trade-off, with an exact deterministic fallback.
 - **Risk model** — interpretable hand-weighted factors, not learned from a labelled dataset.
 - **Evaluation** — a canonical-case demonstration (3 cases + 3 controls + an ablation), **not** a population-scale labelled benchmark.
 - **Data** — the ethnobotany source skews to English-documented knowledge, under-representing some communities.
 - **Deployment** — hardened for local/single-user (rate limiting, configurable CORS, bounded inputs, path-safe static serving) but ****no authentication / multi-tenancy**** — the gate for shared public hosting.
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13. Roadmap

Tier 1 — done. Security hardening (path traversal, rate limiting, CORS, input bounds), CI, one-command Docker demo, test-path fix, documentation sync.

Tier 2 — done. Claim-level novelty assessment in the examiner flow (independent/dependent claim splitting, per-claim verdicts, injection-immune).

Tier 3 — strong differentiators (next).

- A larger, expert-labelled evaluation benchmark (precision/recall at scale).
- Full-text and non-US patent coverage; multilingual TK ingestion at scale.
- A live hosted demo URL so reviewers can click rather than build.
- Continuous monitoring + alerting for newly-filed patents.

Moonshot.

- Multi-tenant institutional deployment (auth, per-org isolation).
 - Community consent and governance as a first-class feature.
 - Claim-element (sub-claim) decomposition for finer anticipation analysis.
-

14. How it was built — engineering process

TK-Shield was built **AI-assisted, under direction and review**. The architecture, data-source, evaluation, security, and product decisions are the author's — keyless/offline-first design, hybrid retrieval over pure embeddings, the interpretable risk model, the three-persona structure, the community-attribution

feature, and the WIPO policy framing. Implementation was accelerated with an AI coding assistant.

The work proceeded in disciplined, reviewable increments:

- **Core platform** — search, risk, enrichment, RAG, API, frontend.
- **Scale-up** — real keyless data (PatentsView bulk), registry to ~2k entries, corpus-integrity guarantees.
- **Evaluation & framing** — the landmark harness, honest metrics, the WIPO brief.
- **Audit-driven hardening (Tier 1)** — an engineering-board review found a real path-traversal vulnerability and several productization gaps; all were fixed, tested, and merged via PR with green CI.
- **Feature deepening (Tier 2)** — claim-level novelty, the feature that closes the biggest credibility gap for the examiner audience.
- **Final audit** — full end-to-end verification: 121+22 tests green, every endpoint smoke-tested live, security re-verified, repository confirmed clean and in sync with GitHub.

Every change landed through a pull request with passing CI; the repository tracks no build artifacts or scratch files.

15. Author and attribution

Designed and directed by **Ansul Suryawanshi**, applying an AI/ML and environmental-science background to the defensive protection of traditional knowledge.

Source code — MIT Licence. **Patent metadata** — PatentsView (USPTO, public domain). **TK entries** — Dr. Duke's databases (USDA, CC0) and Wikidata (CC0).

Enrichment — PubMed, Wikidata, GBIF. TK-Shield is a defensive research tool; it surfaces and organizes public evidence and does not constitute legal advice.

Built as a demonstration of how open data and open models can serve the defensive-protection goals of the WIPO IGC and the 2024 Treaty on Genetic Resources and Associated Traditional Knowledge.

Appendix A — repository map

api/	FastAPI app, cached deps, request schemas, routers
src/	
search/	vector_store - keyword_search - hybrid_ranker (RRF)
classifier/	ip_risk_scorer (5-factor) - domain
nlp/	preprocessor - ner_extractor
clients/	pubmed - wikidata - gbif - patentsview - _http (resilient)
enrichment/	prior_art (concurrent fan-out, citation-tagged)
rag/	retriever - report_generator - novelty - claim_parser - llm_client
registry/	tk_store (SQLite + vector collection)
ingestion/	sources/ - tk_sources/ - build_corpus - ingest_to_chromadb - seed_*
evaluation/	landmark_eval
utils/	config (single singleton) - dates - lexicons
frontend/	Vite + React + TS SPA (api - lib - components - features)
docs/	this report - whitepaper PDF + builder - evaluation report
tests/	121 backend tests (network-free)

Appendix B — API reference

Purpose		
Status + LLM / live-patent availability		
TK registry CRUD + paginated search		
Fast hybrid search + 5-factor risk (no LLM/network)		
markdown\	pdf`	Full RAG rep
Examiner claim-level novelty verdict		
Live PatentsView check (degrades gracefully without a key)		
Researcher analytics over registry + corpus		

Appendix C — reproduce everything

```
# Fastest – instant landmark demo (no keys, no big download)
docker compose up --build # -> http://localhost:8000

# Tests
pytest tests/ # 121 backend, network-free
npm --prefix frontend run test # 22 frontend

# Evaluation (writes docs/evaluation_report.{md,json})
PYTHONPATH=. python -m src.evaluation.landmark_eval

# Rebuild this report's PDF from the markdown
PYTHONPATH=. python docs/build_project_report.py # -> docs/TK-Shield-Project-Report.pdf
```